

SHARUN KUHAR

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EDUCATION

- **Johns Hopkins University** Sep. 2020 - Apr. 2025
Ph.D., Mechanical Engineering Baltimore, MD, USA
Dissertation: Computational fluid dynamics based models of gastric digestion in health and disease
Advisor: Rajat Mittal
- **Indian Institute of Technology, Kanpur** 2014 - 2019
B.T.–M.T. Dual Degree, Mechanical Engineering Kanpur, UP, India
GPA: B.Tech. – 9.9/10, M.Tech. – 10/10
Dissertation: DNS and Stability Analysis of Flow Past a Square Cylinder Placed in a Channel
Advisor: Arun K. Saha

HONORS AND AWARDS

- **Creel Family Teaching Assistant Award** (2025): Best teaching assistants in the department [\[link\]](#)
- **Quad Fellowship** (2023): Cohort of 100 graduate students in STEM from Australia, India, Japan, and the USA who demonstrate academic excellence and passion for societal impact [\[link\]](#)
- **General Proficiency Medal** (2019): Graduated at the top of the department [\[link\]](#)
- **O.P. Bajaj Memorial Award** (2017): Best final year undergraduate in the department [\[link\]](#)
- **Academic Excellence Awards** (2018, 2016, 2015) [\[link\]](#)
- **Summer Undergraduate Research Grant for Excellence (SURGE)** (2016) [\[link\]](#)

RESEARCH EXPERIENCE

- **Injectable Medicines** May 2025 - Present
Postdoctoral Research, Purdue University (Advisor: Arezoo Ardekani)
 - Building predictive models of injectable medicines to reduce risks associated with drug development and predict the outcome of new therapies in humans
 - Developed finite-element based poroelasticity solver to model tissue mechanics
 - Coupled the tissue model with an absorption model to predict systemic delivery and bioavailability [\[p11\]](#)
- **Stomach Biomechanics** Sep. 2020 - Apr. 2025
PhD Research, Johns Hopkins University (Advisor: Rajat Mittal)
 - Development:**
 - Developed an imaging data-based computational model of the stomach using an immersed boundary solver
 - Incorporated chemical reactions of food hydrolysis to perform chemo-fluid dynamic simulations [\[p3\]](#) [\[cp1\]](#)
 - Employed particle resolved simulations to model tablets and large food particles [\[p9\]](#) [\[p2\]](#)
 - Implemented Lagrangian Point Particle Model (LPPM) to simulate small sized food particles
 - Used Volume of Fluids (VOF) approach to account for different density fluids [\[p6\]](#) [\[p7\]](#)
 - Application:**
 - Studied the effect of posture and motility disorders on the dissolution of an oral tablet [\[p2\]](#)
 - Quantified the effects of motility disorders on the mixing and hydrolysis function of the stomach [\[p3\]](#) [\[cp1\]](#)
 - First-of-its-kind tool to assess the efficacy of pyloric surgery in gastroparesis patients [\[p5\]](#)
 - Studied the mechanism of gastritis due to bile reflux in patients [\[p6\]](#)
 - Captured the forces responsible for the breakdown of large solid food particles inside the stomach [\[p9\]](#)
 - Studied the survival of pathogens inside the stomach leading to food poisoning [\[p10\]](#)

- Fiber Optic Cable Design** 2019 - 2020
 Modeling & Simulations Division, R&D Department, Sterlite Technologies, Dadra & Nagar Haveli
 - Developed theoretical and semi-empirical models to predict the onset of signal attenuation in fiber optic cable designs under crush force and winding tension
 - Simulated structural deformation in finite-element software to analyze cable designs
 - Developed reduced-order model based tools for predicting fiber-optic cable response in extreme temperatures
 - Patented grooved cable designs with higher drag to enable air blowing cables to longer distances [pat2]
 - Patented flexible ribbon design to enable more efficient packing of fibers inside the cable [pat1]
- Hydrodynamic Instability** 2018 - 2019
 Masters Thesis, IIT Kanpur (Advisor: Arun K. Saha)
 - Studied the flow past a square cylinder at different blockages using direct numerical simulations
 - Developed a code to perform the linear stability analysis of steady flow in the same configuration
 - Studied the effect of wall proximity on transition from steady to unsteady (Hopf bifurcation) and from symmetric to asymmetric flow (Pitchfork bifurcation) [pre1]
- Non-Newtonian Fluids** May - Jul. 2017
 MITACS Globalink Summer Internship, McGill University (Advisor: Agus P. Sasmito)
 - Studied the flow of mine backfill slurry (40-70% solid fraction) through the hydraulic network in mines
 - Modeled the slurry as a two-phase mixture, using Dense Discrete Phase Model (DDPM), and as a Bingham fluid to predict the pumping requirements through different approaches
 - Compared the simulations against different friction factor model predictions and against in-situ data [p1]

JOURNAL PUBLICATIONS

- [p11] **S. Kuhar**, C. Li, and A. Ardekani, “Multiscale modeling of subcutaneous administration of peptides”, *bioRxiv* 2026.06.01.724102 (2026). [doi]
- [p10] **S. Kuhar**, J. H. Seo, and R. Mittal, “Flow-mediated regulation of pathogen survival in the human stomach”, *arXiv:2606.10895* (2026). [doi]
- [p9] **S. Kuhar**, J. H. Seo, and R. Mittal, “*In-silico* study of the forces causing solid food breakdown inside the stomach”, *Journal of the Royal Society Interface*, 22.229 (2025): 20250291. [doi]
- [p8] W. Li, **S. Kuhar**, J. H. Seo, and R. Mittal, “Computational Study of Gastric Emptying in Microgravity: The Role of Density and Viscosity Variations”, *Physics of Fluids*, 37.7 (2025). [doi]
- [p7] W. Li, **S. Kuhar**, J. H. Seo, and R. Mittal, “Modeling the Effect of Sleeve Gastrectomy on Gastric Digestion in Stomach: Insights from Multiphase Flow Modeling”, *Journal of Biomechanical Engineering*, 147.6 (2025). [doi]
- [p6] **S. Kuhar**, J. H. Seo, M. Camilleri, P. J. Pasricha, and R. Mittal, “Duodenogastric Reflux in Health and Disease: Insights from a Computational Fluid Dynamics Model of the Stomach”, *American Journal of Physiology-Gastrointestinal and Liver Physiology*, 328.4 (2025): G411-G425. [doi]
- [p5] **S. Kuhar**, J. H. Seo, P. J. Pasricha, and R. Mittal, “In silico modelling of the effect of pyloric intervention procedures on gastric flow and emptying in a stomach with gastroparesis”, *Journal of the Royal Society Interface*, 21(210), (2024). [doi]
- [p4] **S. Kuhar** and R. Mittal, “Computational Models of the Fluid Mechanics of the Stomach.” *Journal of the Indian Institute of Science*, 104.1 (2024): 65-76. (Invited Review Article) [doi]
- [p3] **S. Kuhar**, J. H. Lee, J. H. Seo, P. J. Pasricha, and R. Mittal, “Effect of stomach motility on food hydrolysis and gastric emptying: Insight from computational models”, *Physics of Fluids*, 34(11), 111909, (2022). [doi]

- [p2] J. H. Lee, **S. Kuhar**, J. H. Seo, P. J. Pasricha, and R. Mittal, “Computational modeling of drug dissolution in the human stomach: Effects of posture and gastroparesis on drug bioavailability”, *Physics of Fluids*, 34(8), 081904, (2022). [doi]
- [p1] B. Bharathan, M. McGuinness, **S. Kuhar**, M. Kermani, F. P. Hassani, and A. P. Sasmito, “Pressure loss and friction factor in non-Newtonian mine paste backfill: Modelling, loop test and mine field data.” *Powder Technology*, 344, 443–453, (2019). [doi]

CONFERENCE PAPERS

- [cp1] **S. Kuhar**, J. H. Seo, P. J. Pasricha, and R. Mittal, “Computational Fluid Dynamics of Digestion Inside the Stomach.” *Proceedings of the 10th International and 50th National Conference on Fluid Mechanics and Fluid Power (FMFP)*, BFM-057 (2023). [pdf]

CONFERENCE PRESENTATIONS

- [pre15] **S. Kuhar**, J. H. Seo, P. Pasricha, and R. Mittal, “Digital Twin of the Stomach: Applications to Digestive Disorders and Surgery Planning” *American Physiology Summit*, (2025). [link]
- [pre14] **S. Kuhar**, J. H. Seo, and R. Mittal, “Towards a Digital Twin of the Stomach with Application to Digestion and Gastric Reflux” *APS Division of Fluid Dynamics*, L04.00005 (2024). [link]
- [pre13] W. Li, **S. Kuhar**, J. H. Seo, and R. Mittal, “Food Digestion in the Stomach after Bariatric Surgery: Insights from Multiphase Flow Modeling” *APS Division of Fluid Dynamics*, L04.00007 (2024). [link]
- [pre12] **S. Kuhar**, A. Menys, J. H. Seo, and R. Mittal, “Computational Modeling of Solid Food Digestion Inside the Stomach.” *APS Division of Fluid Dynamics*, L10.00008 (2023). [link]
- [pre11] **S. Kuhar** and R. Mittal, “Computational Modeling of Digestion and Drug-dissolution Inside the Stomach.” *ReCoVor*, 64A, (2023). [link]
- [pre10] **S. Kuhar**, A. Menys, J. H. Seo, and R. Mittal, “StomachSim: An in-silico model of stomach biomechanics based on patient-specific imaging data.” *NeuroGASTRO-2023*, (2023). [link]
- [pre9] **S. Kuhar**, J. H. Seo, P. J. Pasricha, and R. Mittal, “StomachSim: An in-silico simulator of gastric biomechanics with application to pyloroplasty.” *American Physiology Summit*, 38, 5729950 (2023). [link]
- [pre8] **S. Kuhar**, J. H. Seo, P. J. Pasricha, and R. Mittal, “StomachSim: An in-silico simulator of gastric biomechanics with application to pyloroplasty.” *Digesta Disease Week*, 164.6 (2023). [link]
- [pre7] **S. Kuhar**, J. H. Lee, J. H. Seo, P. J. Pasricha, and R. Mittal, “StomachSim: an in-silico simulator of gastric biomechanics” *The Johns Hopkins Department of Medicine & Whiting School of Engineering Research Retreat*, (2023).
- [pre6] **S. Kuhar**, J. H. Lee, J. H. Seo, P. J. Pasricha, and R. Mittal, “Biofluid dynamics of digestion in the stomach: Insights from computational modeling.” *APS Division of Fluid Dynamics*, Z07.00004 (2022). [link]
- [pre5] **S. Kuhar**, J. H. Lee, J. H. Seo, P. J. Pasricha, and R. Mittal, “Biofluid dynamics of digestion in the stomach: Insights from computational modeling.” *The Johns Hopkins Department of Medicine & Whiting School of Engineering Research Retreat*, (2022).
- [pre4] **S. Kuhar**, J. H. Lee, J. H. Seo, P. J. Pasricha, and R. Mittal, “The Chemo-Fluid Dynamics of Digestion in the Stomach: Insights from Computational Modeling.” *CEAFM Burgers Symposium*, (2022). [link]

- [pre3] **S. Kuhar**, J. H. Lee, J. H. Seo, P. J. Pasricha, and R. Mittal, “The Chemo-Fluid Dynamics of Digestion in the Stomach: Insights from Computational Modeling”, *APS Division of Fluid Dynamics*, T15.00006, (2021). [\[link\]](#)
- [pre2] J. H. Lee, **S. Kuhar**, J. H. Seo, P. J. Pasricha, and R. Mittal, “The Fluid Dynamics of the Dissolution of an Oral Drug in the Human Stomach”, *APS Division of Fluid Dynamics*, T15.00007, (2021). [\[link\]](#)
- [pre1] **S. Kuhar**, Arun K. Saha, “Linear Stability Analysis Of Two-Dimensional Flow Past A Square Cylinder At Different Blockage Ratios”, *Research Scholar Day*, Association of Mechanical Engineers, IITK, FTS-18, (2019). [\[link\]](#)

PATENTS

- [pat2] **S. Kuhar**, V. Shukla, S. Sharma, and K. Sahoo, Sterlite Technologies Ltd, 2021. “Ribbed and grooved cable having embedded strength member with water blocking coating”. U.S. Patent Application 17/347,080 (Granted: 2023). [\[link\]](#)
- [pat1] H. Kondapalli, S. Sharma, **S. Kuhar**, A. Nath, V. Shukla, and B. Sarkaar, Sterlite Technologies Ltd, 2021. “Intermittently bonded optical fibre ribbon with unequal bond and gap lengths”. U.S. Patent Application 17/139,508 (Granted: 2023). [\[link\]](#)

MENTORING

- Daira Sofia Velasco Vega, PhD Student, Purdue University 2025 - Present
- Rushikesh Prafulla Deotale, PhD Student, Purdue University May 2026 - Present
- Weixuan Li, Master’s Student, Johns Hopkins University 2023 - 2025
- Aditi Gupta, Undergraduate Student, Johns Hopkins University May - Jul. 2023

TEACHING EXPERIENCE

- Teaching Assistant at Johns Hopkins University:
 - Numerical Methods, Fall 2021 and Fall 2023
 - Computational Fluid Dynamics, Spring 2022 and Spring 2024
- Teaching Assistant at IIT Kanpur:
 - Numerical Methods, Fall 2018
 - Turbulent Flow, Spring 2019

SERVICE AND OUTREACH

- **Co-organizer**, Mini-symposium 6.5: “Computational Modeling of Physiological Flows” 2026
Society of Engineering Science (SES) Annual Technical Meeting, Purdue University
 Co-organizers: Arezoo Ardekani, Dongjie (Fred) Jia
- **Core Team Member** 2016 - 2017
 Institute Counseling Service, IIT Kanpur
 - Part of a 10-member team responsible for campus-wide counseling service activities
 - Negotiated with banks to raise 150k INR in scholarship for needy students
 - Led a team of 137 student-guides during the freshmen orientation program with 450k INR budget
 - Worked with professional counselors to provide emotional, mental, and financial support to students